

# The Sugar Trap

*Identifying Market Gaps in High-Protein, Low-Sugar*

*Snacking*

**Executive Data Analysis | Open Food Facts**

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# Executive Summary



## The Market Gap

A significant "empty quadrant" exists for high-protein products with < 10g sugar, particularly in the Baked Goods category.



## Formulation Insight

Text analysis identified Soy, Whey, and Pea/Peanut as the dominant protein sources for existing success stories.



## Strategic Pivot

Leveraging high-fiber ingredients can further differentiate new entries in the health-conscious snack segment.

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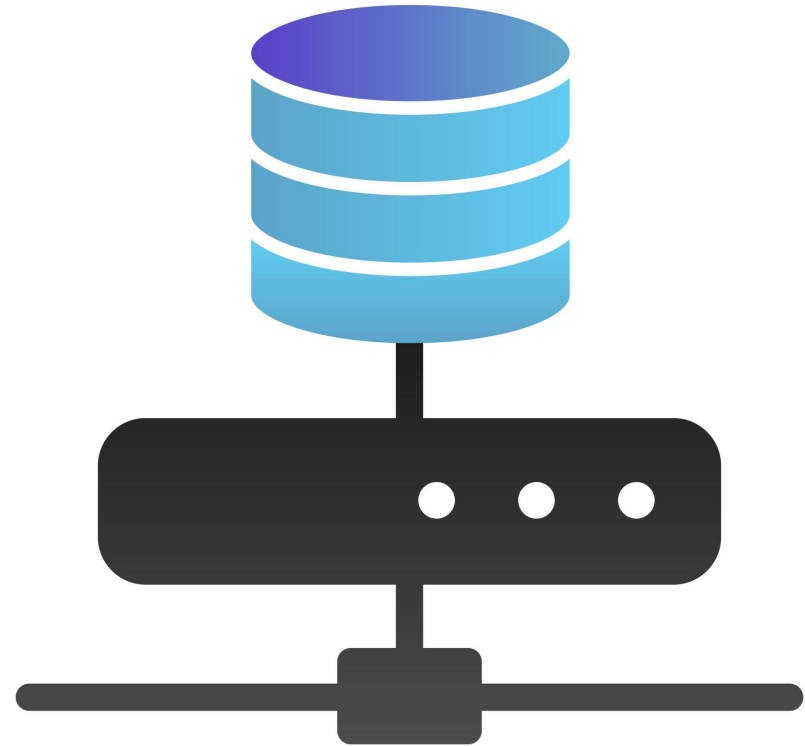
# Part I: Methodology

Turning 3GB of Raw Data into Business Intelligence

# Data Engineering Pipeline

Successfully processed the Open Food Facts repository using a custom ingestion pipeline designed for high memory efficiency.

- **Ingestion:** Python-based chunking of 3GB CSV.
- **Subsetting:** Focused on 500k high-impact rows.
- **Memory Management:** Optimized ``usecols`` loading.
- **Deployment:** Interactive dashboard via Streamlit.



# | Addressing the "Data Traps"

## **Biological Consistency**

Removed outliers where sugar or protein values exceeded 100g per 100g serving—ensuring physical reality.

## **Missing Metric Handling**

Dropped records missing product names or core nutritional markers to maintain the integrity of the Matrix visualization.



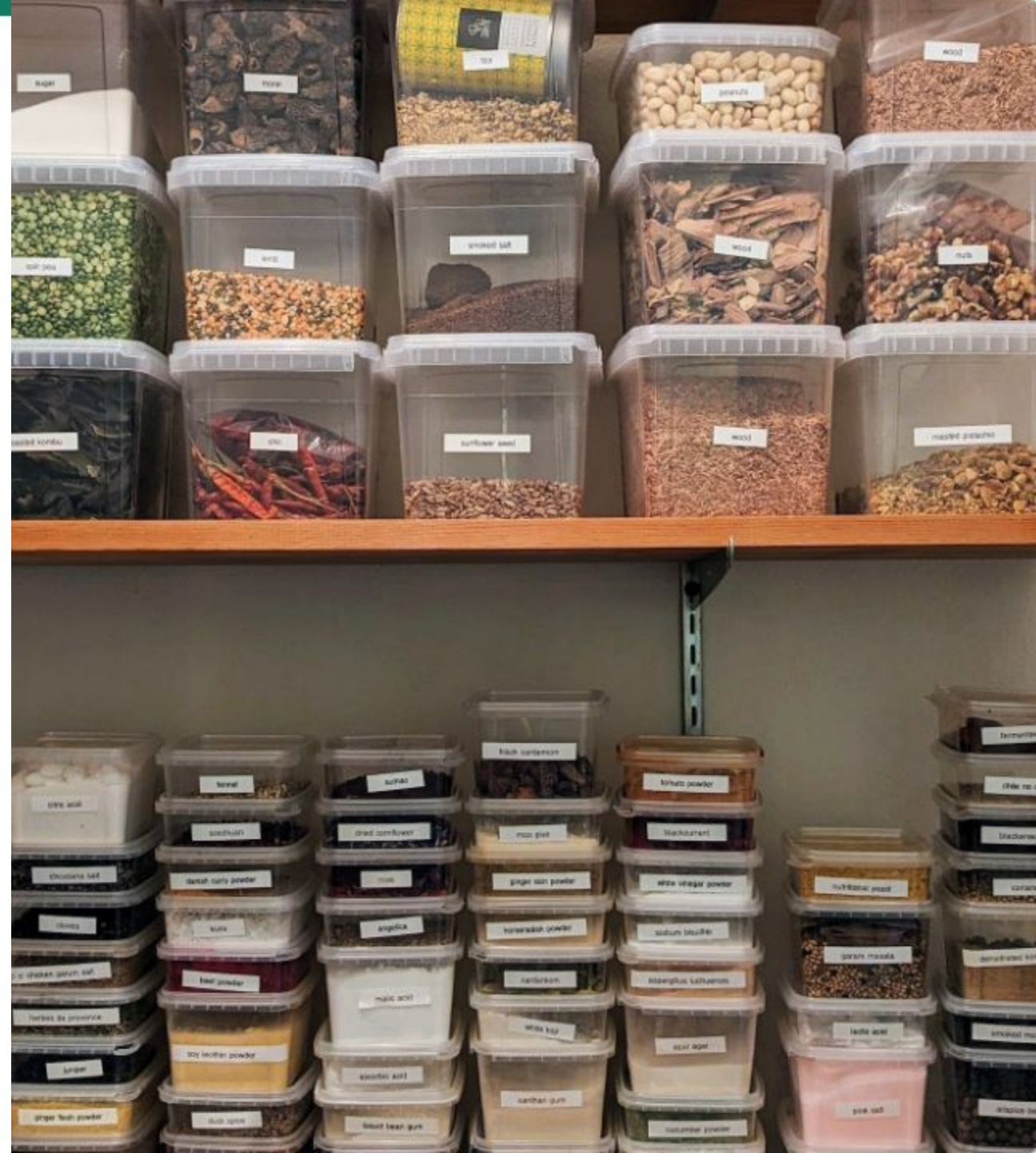
# Part II: Market Mapping

Visualizing Competition and Opportunity

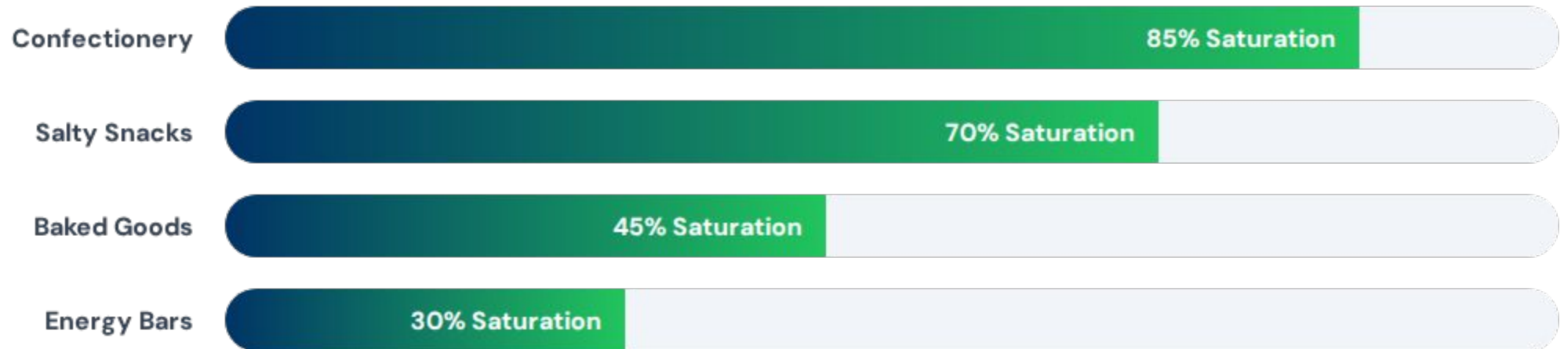
# Category Bucketing

Chaos to Clarity: We mapped over 5,000 messy nutritional tags into 5 strategic business units.

- ✓ Confectionery
- ✓ Salty Snacks
- ✓ Energy Bars
- ✓ Baked Goods



# Snack Market Distribution



*Note: Lower saturation in health-focused categories indicates high growth potential.*



# Scale of Analysis

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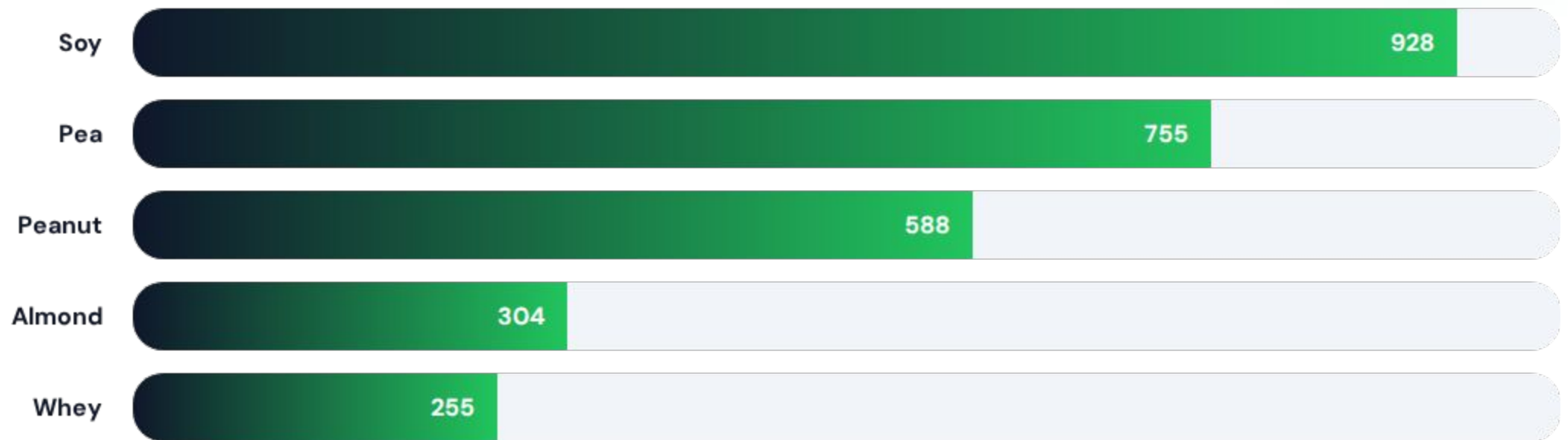
**4,651**  
**High-Protein Products Analyzed**

## **Substantial Data Evidence**

Our analysis focused on the high-protein segment (>15g per 100g serving). This robust sample size ensures the reliability of our ingredient and market gap findings.

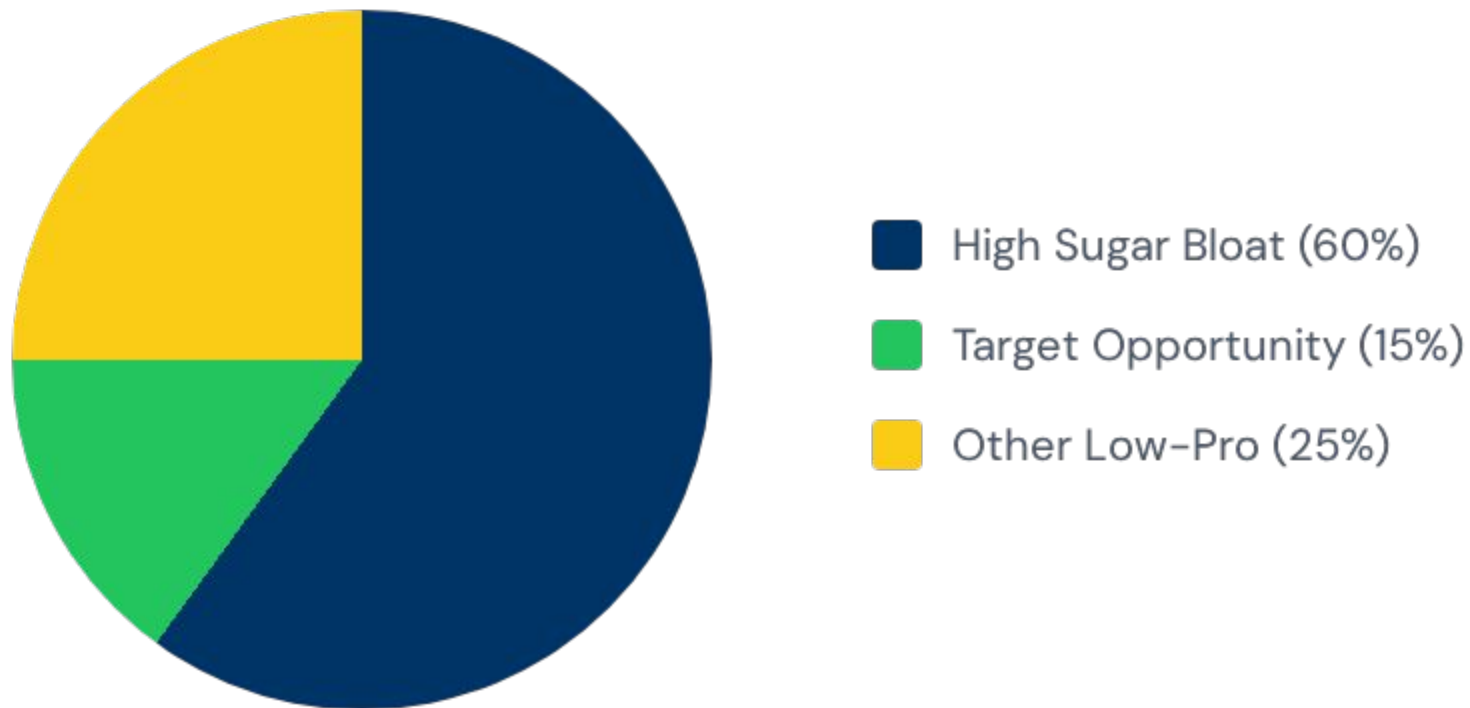
# Top Protein Sources (Keyword Frequency)

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*Source: Text analysis of ingredient strings in 4,651 high-protein products.*

# | The Target "Market Gap"



The target segment (>15g Protein, <10g Sugar) currently occupies only 15% of the total dataset.

# Ingredient Analysis: Top Sources



## Soy Protein

Most common plant-based source found in Energy Bars.



## Pea Protein

Primary choice for high-bioavailability fitness snacks.



## Peanut

Fastest-growing allergen-friendly protein alternative.

## | Final Recommendation

*"The market is saturated with high-sugar confectionery. The real 'Market Gap' lies in **Low-Sugar Baked Goods** utilizing **Soy and Pea protein** blends."*

— *Strategic Growth Plan*

# Questions?

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Interactive dashboard and code available on GitHub.

**Data Engineering Take-Home Lab Portfolio**